Experiment - 1

Aim:

To design an Entity-Relationship (ER) diagram for an Indian e-commerce platform, identifying all entities, attributes, primary keys, composite & multivalued attributes, weak entities, specialization of product and participation constraints.

Source code (Design specification & ER Notation)

1. Entities, Primary Keys (PK), Composite & Multivalued attributes.

Entity	Primary Key	Composite Attributes	Multi-valued Attributes
Customer	customer_id	Name (First_Name, Middle_Name, Last_Name)	Phone_No, Email_id
Address	address_id	House_No, Street, city, State, Pin_code	—
Seller	seller_id	–	Phone_No
Category	category_id	–	—
Product	product_id	–	Keywords
Order	order_id	–	—
OrderItem	(Weak Entity)	–	–
Payment	payment_id	–	–
Delivery	delivery_id	–	–
Admin	—	—	Email_id

2. Weak Entity

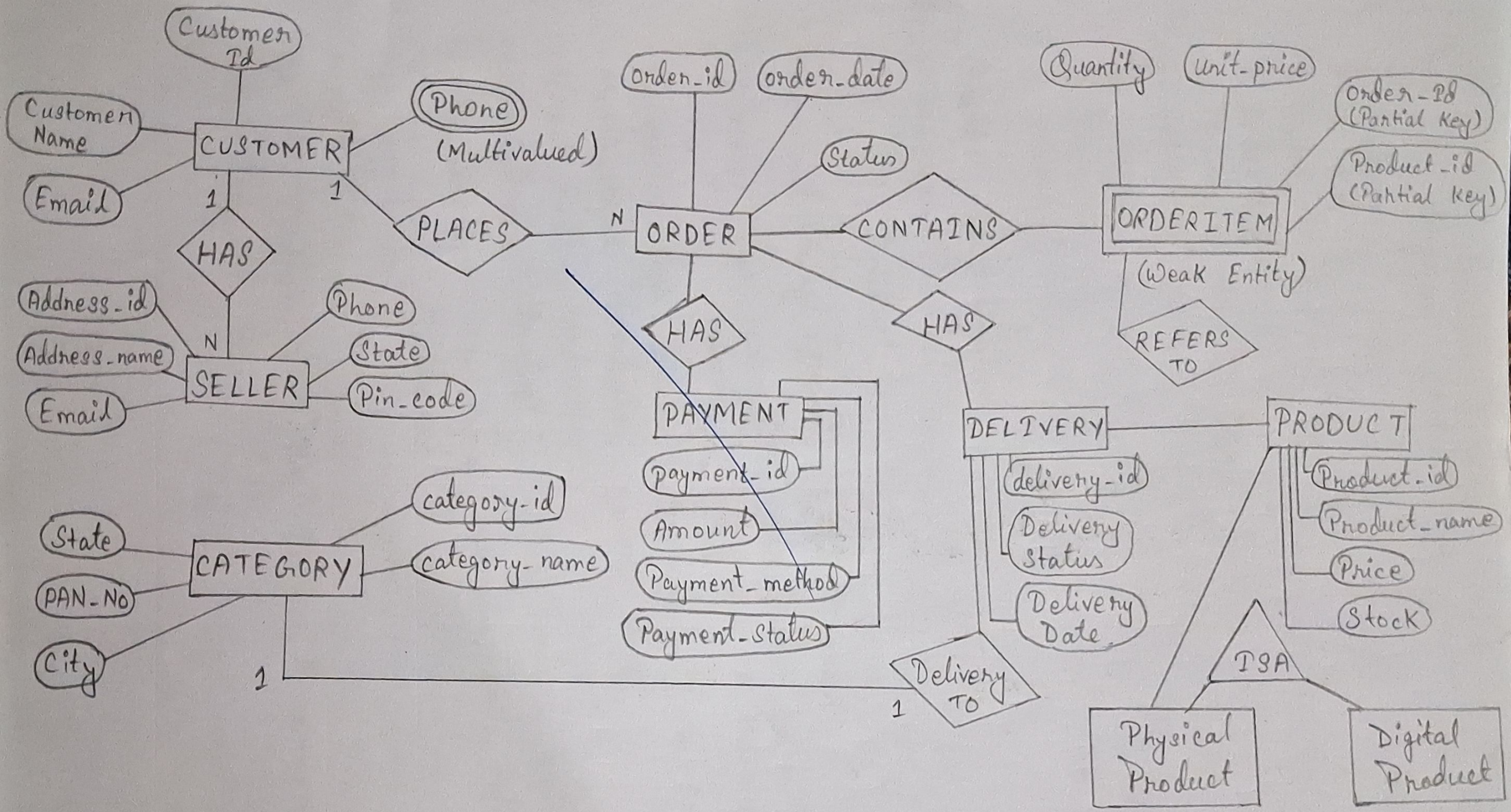
OrderItem is a weak entity because it cannot exist independently without an Order. Its discriminator (Partial Key) is item_seq (a serial number within the order). Its complete primary key is (order_id, item_seq) where order_id is derived from the identifying

relationship contains with the Order entity.

3. Specialization of Product (Disjoint & Total)
The Product entity is specialized into three subclasses. This is a disjoint specialization (a product belongs to exactly one subclass) and total (every product must fall into one of these catagories).
- Electronics → Attributes: Warrenty-Period, Power-Rating, Brand.
- Apparel → Attributes: Size, Material, Gender
- Grocery → Attributes: Expiry-Date, Weight, Nutritional-Info

4. Relationships & Participation Constraints

Relationship	Participating Entities	Mapping	Participation Constraints
Places	customer - Order	1 : N	Total participation of Order (every order must be placed by a customer)
Has-address	customer-address	1 : N	Total participation of Address.
Sells	Seller - Product	M : N	Total participation of Product. Total participation of seller.
Belongs-To	Product - Category	N : 1	Total participation of Product.
Contains	Order - OrderItem	1 : N	Total participation of OrderItem
Includes	OrderItem - Product	N : 1	Total participation of OrderItem
Makes-Payment	Order - Payment	1 : 1	Total participation of Payments
Has-Delivery	Order - Delivery	1 : 1	Total participation of Delivery.


Customer Id
Customer Name
Email
CUSTOMER
Phone
(Multivalued)
1
1
HAS
PLACES
N
ORDER
order_id
order_date
Status
CONTAINS
Quantity
Unit_price
Order_ID (Partial Key)
Product_id (Partial key)
ORDERITEM
(Weak Entity)
REFERS TO
Address_id
Address_name
Email
N
SELLER
Phone
State
Pin_code
HAS
HAS
PAYMENT
Payment_id
Amount
Payment_method
Payment_status
DELIVERY
delivery_id
Delivery Status
Delivery Date
PRODUCT
Product_id
Product_name
Price
Stock
State
PAN_No
City
CATEGORY
category_id
category_name
1
1
Delivery TO
ISA
Physical Product
Digital Product

ER Diagram

The ER diagram contains the following entities

1. Customer
2. Address
3. Order
4. OrderItem
5. Product
6. Seller
7. Category
8. Payment
9. Delivery

Product Specialization

1) Physical Product
2) Digital Product

Important things shown in ER Diagram

- Primary Keys: customer-ID, product-ID, Order-ID, seller-ID, category-ID, Payment-ID, Delivery-ID
- Composite Keys: order-ID, + product-ID in OrderItem
- Multi-valued Attribute: customer phone
- Weak Entity: orderItem
- Specialization Product: Physical Product, Digital Product
- Relationships: Places, has, contains, sells, Refers-To, Delivered-To

Result

The ER Diagram for the Indian E-commerce platform was designed successfully.